

CSE 8803: Deep Learning for Text Data

Fall 2024

Homework 3

Due: Nov.5, 2024

1. $\text{\LaTeX}$ Instructions

If you opt to use the provided $\text{\LaTeX}$ template, employ the command “ $\text{\textbackslash answer\{}}$ ” to insert your answers. Placeholders are already set up for you. Simply place your answers inside the curly brackets.

2. (40 points) Reinforcement Learning from Human Feedback (RLHF) with Proximal Proxy Optimization (PPO)

You are aiming to align a large language model to human values, ensuring that its outputs are helpful and harmless. The key components are 1) learning a reward model from human preference and 2) optimizing a policy against the reward model. We will go through them in this problem.

Part A: Reward Modeling

To train our reward model, we can start from a supervised baseline and add a randomly initialized linear head that outputs a scalar value [3, 1]. The loss function for the reward model is:

$$\text{loss}(\theta) = -\mathbb{E}_{(x, y_u, y_l) \sim D} [\log(\sigma(r_\theta(x, y_u) - r_\theta(x, y_l)))], \quad (1)$$

where $r_\theta(x, y)$ is the scalar output of the reward model for prompt x and completion y with parameters θ , y_u is the preferred completion out of the pair of y_u and y_l , and D is the dataset of human preference.

Consider the following dataset for reward modeling:

Prompt	Model's Response A	Model's Response B
P1	Lending a hand can light up someone's day.	Turn a blind eye when others are struggling.
P2	Keeping wisdom to oneself is a way to power.	Sharing insights enriches the community.
P3	Withholding guidance does no harm.	Guiding others on the right path can change their future.

Table 1. Reward Modeling Dataset.

Assume that for each prompt and response pair, the reward model assigns the following scalar outputs:

Prompt	$r(P, \text{ResponseA})$	$r(P, \text{ResponseB})$
P1	0.9	-
P2	0.3	-
P3	0.4	-

Table 2. Incomplete Reward Values.

- (a) (5 points) Please complete the above missing reward values using $\{0.1, 0.8\}$.

[Answer: Please refer to table 3 below:]

Prompt	$r(P, \text{ResponseA})$	$r(P, \text{ResponseB})$
P1	0.9	0.1
P2	0.3	0.8
P3	0.4	0.8

Table 3. Completed Reward Values

- (b) (10 points) Calculate the individual loss for each prompt and the overall loss for the entire dataset.

[Answer:

$$\text{loss}_{P1} = -\log(\sigma(0.9 - 0.1)) = -\log(0.690) = 0.37$$

$$\text{loss}_{P2} = -\log(\sigma(0.8 - 0.3)) = -\log(0.622) = 0.47$$

$$\text{loss}_{P3} = -\log(\sigma(0.8 - 0.4)) = -\log(0.599) = 0.51$$

$$\text{Overall Loss} = \frac{\text{loss}_{P1} + \text{loss}_{P2} + \text{loss}_{P3}}{3} = 0.45$$

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Part B: RL with PPO

With the reward model, we can train a policy that generates higher-quality outputs aligned with human values. To achieve this, you're employing PPO [2] as your primary reinforcement learning algorithm and incorporating human feedback to guide the model's behavior.

Different from the above reward modeling, we now have a new full reward R defined as:

$$R(x, y) = r_\theta(x, y) - \beta \log \left(\frac{\pi_\phi^{RL}(y|x)}{\pi^{SFT}(y|x)} \right), \quad (2)$$

where $\beta = 0.5$ is the KL reward coefficient, π_ϕ^{RL} is the current RL policy, and π^{SFT} is the original SFT model.

- (a) (5 points) You may find that $R(x, y)$ include a term in the reward that penalizes the KL divergence between the learned RL policy π_ϕ^{RL} with parameters ϕ and the original supervised model π^{SFT} . Please explain the purpose of the KL term.

[Answer: The KL divergence term in the reward function penalize the RL policy π_ϕ^{RL} for deviating from the supervised model π^{SFT} . The purpose is to align with the original model while maintaining supervised baseline. Therefore, it balances new behavior and human values alignment.]

- (b) (5 points) Given a context x "How to maintain good mental health", the policy (chatbot) generated two responses:

Response A: "10 Practices to Boost Your Mental Well-being: ...";

Response B: "Why Avoiding Negative People Improves Mental Health? ...".

We have the following probabilities and reward model outputs:

$$\begin{aligned} \pi_\phi^{RL}(A|x) &= 0.55, & \pi_\phi^{RL}(B|x) &= 0.45, \\ \pi^{SFT}(A|x) &= 0.60, & \pi^{SFT}(B|x) &= 0.40, \\ r_\theta(x, A) &= 0.9, & r_\theta(x, B) &= 0.7 \end{aligned} \quad (3)$$

Compute the reward $R(x, y)$ for responses A and B.

[Answer:

$$R(x, y) = r_\theta(x, y) - \beta \log \left(\frac{\pi_\phi^{RL}(y|x)}{\pi^{SFT}(y|x)} \right)$$

$$\beta = 0.5$$

$$r_\theta(x, A) = 0.9, r_\theta(x, B) = 0.7$$

$$\pi_\phi^{RL}(A|x) = 0.55, \pi_\phi^{RL}(B|x) = 0.45$$

$$\pi^{SFT}(A|x) = 0.60, \pi^{SFT}(B|x) = 0.40$$

R(x, A):

$$\text{KL penalty: } \log\left(\frac{0.55}{0.60}\right) = \log(0.917) = -0.087$$

$$\text{KL term: } 0.5 \times (-0.087) = -0.044$$

$$R(x, A) = 0.9 - (-0.0435) = 0.944$$

R(x, B):

$$\text{KL penalty: } \log\left(\frac{0.45}{0.40}\right) = \log(1.125) = 0.118$$

$$\text{KL term: } 0.5 \times 0.118 = 0.059$$

$$R(x, B) = 0.7 - 0.059 = 0.641$$

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- (c) (10 points) The Proximal Policy Optimization (PPO) algorithm aims to limit the policy updates to prevent them from being too drastic. This is done using a clipping mechanism in the objective:

$$L^{PPO}(\phi) = \hat{\mathbb{E}}_t[\min(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)], \quad (4)$$

where $\hat{\mathbb{E}}_t$ denotes the expected value at timestep t , $r_t = \frac{\pi_\phi^{RL}(y|x)}{\pi^{SFT}(y|x)}$ is the probability ratio at timestep t , $\hat{A}_t$ is the estimated advantage at timestep t , which typically represents how much better an action is compared to the average action in state s , ϵ is a hyperparameter defining the clipping range to prevent overly large policy updates.

Compute the PPO objective function value for responses A and B. For simplicity, we use $\hat{A}_t = 0.7, \epsilon = 0.2$ for this exercise.

[Answer:

$$L^{PPO}(\phi) = \hat{\mathbb{E}}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

Step 1: Calculate r_t

Response A:

$$r_{tA} = \frac{\pi_{\phi}^{RL}(A|x)}{\pi^{SFT}(A|x)} = \frac{0.55}{0.60} = 0.917$$

Response B:

$$r_{tB} = \frac{\pi_{\phi}^{RL}(B|x)}{\pi^{SFT}(B|x)} = \frac{0.45}{0.40} = 1.125$$

Step 2: Clipping to r_t

clipping range: $[0.8, 1.2]$

$$\textbf{Response A : } clip_A(r_{tA}, 0.8, 1.2) = 0.9167$$

$$\textbf{Response B : } clip_B(r_{tB}, 0.8, 1.2) = 1.125$$

Step 3: Calculate $L^{PPO}(\phi)$ for A and B

$$L^{PPO}(\phi) = \min \left(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right)$$

$$L^{PPO}(\phi)_A = \min(0.9167 \times 0.7, 0.9167 \times 0.7) = \min(0.6417, 0.6417) = 0.642$$

$$L^{PPO}(\phi)_B = \min(1.125 \times 0.7, 1.125 \times 0.7) = \min(0.7875, 0.7875) = 0.788$$

- (d) (5 points) Based on the PPO objective function values for topics A and B, which topic would the PPO algorithm favor after the policy update? Please briefly illustrate your answer.

[Answer:

$$L^{PPO}(\phi_A) = 0.64 < L^{PPO}(\phi_B) = 0.79$$

Therefore, B would be selected due to higher PPO score, indicating higher adjusted reward.]

3. (60 points) Toy GPT Training

Please check the GitHub repository for this coding assignment: <https://github.com/zoom-wang112358/Assignment-3-toyGPT/tree/main>.

This assignment contains 3 code blocks to complete, each with a 20-point value.

References

- [1] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- [2] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [3] Daniel M Ziegler, Nisan Stiennon, Jeffrey Wu, Tom B Brown, Alec Radford, Dario Amodei, Paul Christiano, and Geoffrey Irving. Fine-tuning language models from human preferences. *arXiv preprint arXiv:1909.08593*, 2019.